

CS 2114 - Data Structures & Software Design

Week 3 day 1

CS2114 Office Hours

TA office hours are in Torg 1070

- Log in to My Digital Hand, add your name and photo!
 - *You should have received an email invite*
- Use My Digital Hand to line up once you arrive to Torg
 - *Describe what you need help with in detail*
- Keep your browser open to MDH and keep your volume on
 - *Pay attention so that you know when it is your turn!*
- CS2114 TAs should be on the far left of the room when you walk in

Class Business

- **Canvas Calendar** – Week 3 Schedule
- **Lab 03** is due this week.
- Work on **Project 1**
- Peer+ credit for week 2 should be visible in canvas
- Optional MC questions most weeks in Modules for extra practice/test prep

Lab 02 - Review

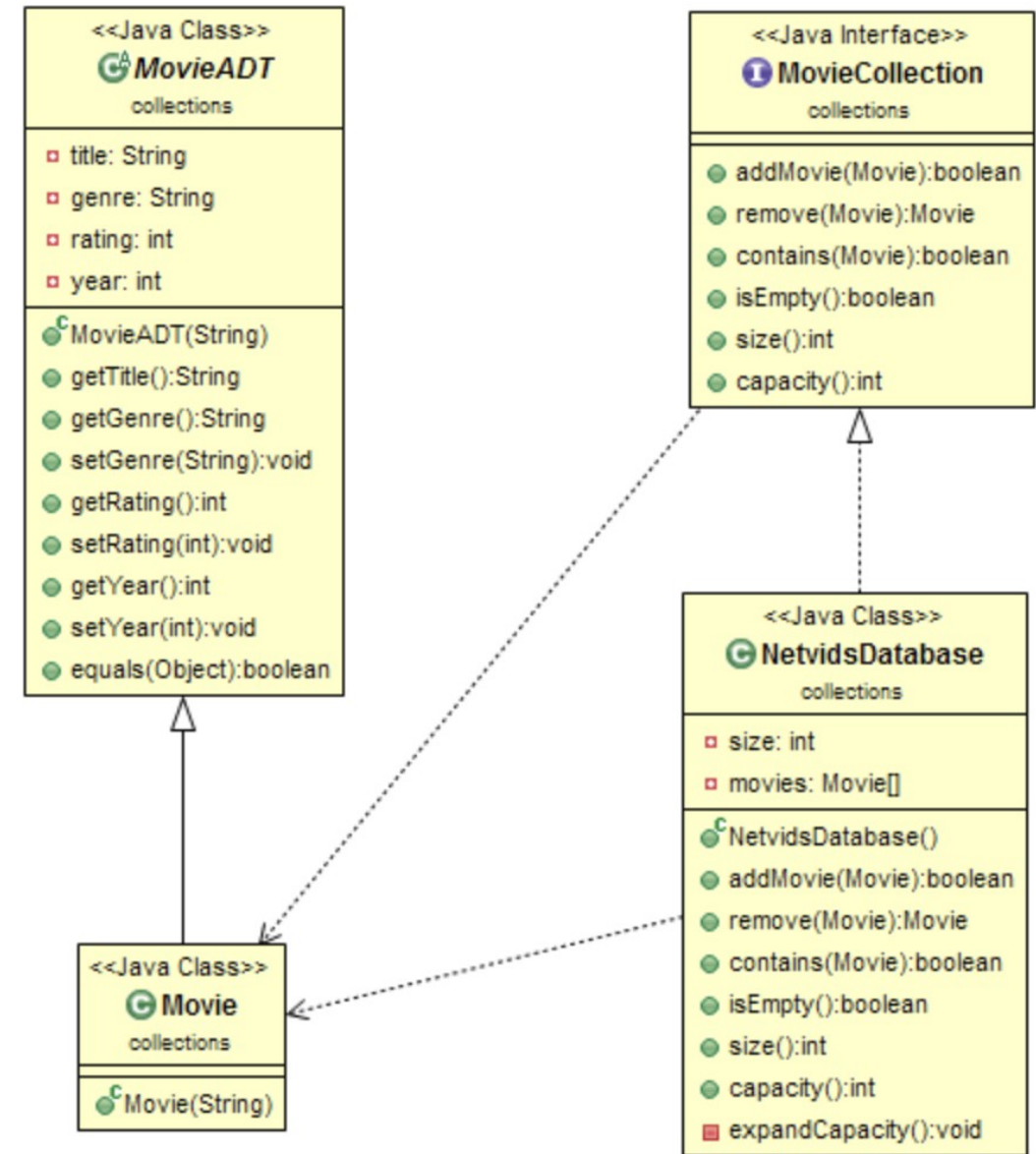
- `Employee emp1 = new Employee(param, param,..);` **Possible or Not Possible?**
- `PartTimeEmployee pt1 = new Employee(param, param,..);` **Possible or Not Possible?**
- `Employee emp2 = new PartTimeEmployee(param, param,..);` **Possible or Not Possible?**

Also, **packages** are folders for grouping similar classes.

Note this can be useful for visibility modifiers... `public`, `private`, `protected`(child classes can access), and default is within the same package.

Lab 03 - Hints

- Abstract Class
- ADT vs Interface



UML Class Diagram

Project 1 Guidance

- Focus on setup of project
- Finish the Bags content pages before starting project 1
- Go through the example code for window class
- Demo setup and output as needed
- **Project 1 instructions reference Testable Random documentation: <https://courses.cs.vt.edu/~cs1114/api/student/TestableRandom.html>**

Why We Use Peer Instruction

- Research shows better student learning and satisfaction from active learning
 - Twice the learning gains in Physics
 - Half grade improvement in Computer Science
- Instructor displays hard multiple-choice question
 - Students vote individually
 - Discuss with peers
 - Students vote individually again
 - Instructor leads a discussion
- Figure out your group of 3-4 students and move around as needed

Tips for a good group discussion

- Take turns being the first one to talk
- Once you all agree on the answer, **don't stop!**
 - Always **go over each wrong answer** and explain why it is wrong
 - Also interesting and useful to think about why somebody might be tempted to choose it—why was that wrong answer included?
 - Even if your group-mate has said something very clearly and correctly, it's a good idea to **repeat it yourself**
 - “So, what I think you said was, ...”
 - **Might seem pointless, but your brain will remember better if YOU say it too**

Week 3 Day 1 Peer +

Q-1: Given the following code:

```
for (int i = 0; i < values.length; i++) {  
    if (values[i] == searchedValue) {  
        found = true;  
    } else {  
        found = false;  
    }  
}  
if (found) {  
    System.out.println("found " + searchedValue);  
} else {  
    System.out.println("not found");  
}
```

If the initial values are:

```
int[] values = { 3, 6, 10, 12, 100, 200, 250, 500 };  
int searchedValue = 250  
boolean found = false;
```

What is the output?

- ☐ A. found 250
- ☐ B. not found
- ☐ C. found 500
- ☐ D. An IndexOutOfBoundsException is thrown

Check Me

Activity: 1 Multiple Choice (vtcs2_arrays_tracing)

Q-1: Given the following code for the ArrayBag class:

```
private T[] contents; // Array where the contents of the Bag are stored
private int numberOfEntries; // How many entries are in the Bag

public int getFrequencyOf(T anEntry)
{
    int count = 0;
    for (int i = 0; i < numberOfEntries; i++)
    {
        if(anEntry.equals(contents[i]))
        {
            count++;
        }
    }
    return count;
}

private int getIndexOf(T anEntry)
{
    for (int i = 0; i < numberOfEntries; i++)
    {
        if (anEntry.equals(contents[i]))
        {
            return i;
        }
    }
    return -1;
}
```

Which of the following 4 implementations could not be used for contains(T) in an ArrayBag?

☐ A.

```
public boolean contains(T anEntry)
{
    boolean found = false;
    int i = 0;
    while ( !found && (i < numberOfEntries))
    {
        if (anEntry.equals(contents[i]))
        {
            found = true;
        }
        i++;
    }
    return found;
}
```

☐ B.

```
public boolean contains(T anEntry)
{
    return getIndexOf(anEntry) > -1;
}
```

☐ C.

```
public boolean contains(T anEntry)
{
    return getFrequencyOf(anEntry) > 0;
}
```

☐ D.

```
public boolean contains(T anEntry)
{
    int i = 0;
    while ((i <= numberOfEntries))
    {
        if (anEntry.equals(contents[i]))
        {
            return true;
        }
        i++;
    }
    return false;
}
```

Python Tutor Demo - Object References

- In eclipse, can download **ObjectReferencesDemo** (use **blue down arrow**)
- Open pythontutor.com, choose Java
- Paste the Employee class in
- Trace code using Next button

CW – Week 3

Code Workouts – Q1

X670: Turn Bag to Set

Code Workouts – Q1

This arrayBag is of a fixed length. It will *not* have the `expandCapacity` method available. Once the bag is full, nothing else can be added. **//SIZE**

How to implement it?

HINT: Which method can help?

Code Workouts – Q1

When invoked the method takes the T parameter anEntry and determines if anEntry already exists in the bag. **// CHECK DUPLICATES**

- If the bag does not already contain anEntry then the method attempts to add it to the bag, and return true.
- If the bag already contained anEntry the method *does not* attempt to add anEntry and returns false.

Remember to review the Bags module, specifically the add() method implementation for Bags as it will be helpful here.

Next Class

- **Lab 03 is** due this week. - Please submit Lab 3 to webCAT using Eclipse and if facing any issues contact any TA. **Post Lab 03** is also due this week.
- Start doing **project 1**.
- Labs are **Mandatory**.
- We will use **piazza heavily**. You can filter to select for Instructor posts so you don't miss any announcements.



Thanks

